Action and Geomery

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Introduction

This note describes action and action principle from a geometrical perspective.

Action

Classical mechanics is deterministic, this implies that the phase space inhabited by the phase points (q, p, t), with phase velocity $\mathbf{u} = (\dot{q}, \dot{p}, 1)$ satisfies

$$\nabla \cdot \mathbf{u} = 0 \tag{I.1}$$

where $\nabla = (\partial_q, \partial_p, \partial_t)$
this implies that there exists a vector field $\boldsymbol{\theta}$ such that

$$\mathbf{u} = -\nabla \times \boldsymbol{\theta} \tag{I.2}$$

The gauge can be chosen so that $\boldsymbol{\theta} \to \boldsymbol{\theta} + \nabla f$ with $f = -\int \theta_p dp$, so momentum component vanishes

$$\boldsymbol{\theta} = (\theta_q, 0, \theta_t) \tag{I.3}$$

using Eq. (I.2) yields

$$u_t = \partial_p \theta_q = 1 \tag{I.4}$$

hence

$$\boldsymbol{\theta} = (p, 0, -\mathcal{H}) \tag{I.5}$$

where we have renamed $\theta_t = -\mathcal{H}$, we shall discover the meaning of $\mathcal{H}$ by substituting (I.5) into (I.2), which gives

$$(\dot{q}, \dot{p}) = (\partial_p \mathcal{H}, -\partial_q \mathcal{H}) \tag{I.6}$$

which are simply the Hamilton's equations. $\mathcal{H}$ is the Hamiltonian which generates the time evolution of the state.

Now consider a trajectory γ and its variation γ', which forms a closed surface Σ in the phase space, where the boundary is

$$\partial\Sigma = \gamma - \gamma' \tag{I.7}$$

the flux through the surface is

$$
\begin{aligned}
\Phi &= -\iint_{\Sigma} \mathbf{u} \cdot d\boldsymbol{\Sigma} \\
&= \iint_{\Sigma} \nabla \times \boldsymbol{\theta} \cdot d\boldsymbol{\Sigma} \\
&= \oint_{\partial\Sigma} \boldsymbol{\theta} \cdot d\boldsymbol{\gamma} \\
&= \int_{\gamma} \boldsymbol{\theta} \cdot d\boldsymbol{\gamma} - \int_{\gamma'} \boldsymbol{\theta} \cdot d\boldsymbol{\gamma} \\
&= \delta \int_{\gamma} \boldsymbol{\theta} \cdot d\boldsymbol{\gamma} \\
&= \delta S
\end{aligned} \tag{I.8}
$$

where we have defined the action as

$$
S = \int_{\gamma} \boldsymbol{\theta} \cdot d\boldsymbol{\gamma} \tag{I.9}
$$

consider

$$
\boldsymbol{\theta} \cdot d\boldsymbol{\gamma} = \boldsymbol{\theta} \cdot \mathbf{u} dt = (p\dot{q} - \mathcal{H}) dt = \mathcal{L} dt \tag{I.10}
$$

where $\mathcal{L}$ is the Lagrangian. Substitute (I.10) into (I.9) yields a more familiar form of the action

$$
S = \int_{\gamma} \mathcal{L} dt \tag{I.11}
$$

If the phase velocity field $\mathbf{u}$ generated by $\mathcal{H}$ is tangential to the trajectory γ at all points along it, we have

$$
\mathbf{u} \cdot d\boldsymbol{\Sigma} = 0 \tag{I.12}
$$

then it follows from Eq. (I.8) that the flux Φ vanishes, or equivalently

$$
\delta S = 0 \tag{I.13}
$$

which is precisely the principle of stationary action.

Thus the principle can be regarded as a direct consequence of system being deterministic (I.1).

Appendix A

Action and classical mechanics can be generalised to systems with more degrees of freedom. The generalisation of definition of action (I.9) is

$$S = \int_\gamma \theta \tag{A.1}$$

where the exterior derivatives

$$d\omega = 0 \tag{A.2}$$

$$\omega = -d\theta \tag{A.3}$$

are the generalisation of Eq. (I.1) and (I.2) respectively. Eq. (A.3) is the Poincaré lemma.

Note the generalised Stokes theorem

$$\int_{\partial\Gamma} \iota = \int_\Gamma d\iota \tag{A.4}$$

where ι is a diffrential form and Γ is a manifold.

In our case, manifold Γ is just a surface and ω is a 2-form,

$$\omega(\mathbf{v}, \mathbf{w}) = \det G(\mathbf{u}, \mathbf{v}, \mathbf{w}) \tag{A.5}$$

where $\mathbf{v}$ and $\mathbf{w}$ are phase velocity vector tangent to trajectory and direction of variation, respectively. The RHS is known as the Gramian which is the signed volume of the parallelepiped spanned by the three vectors, which is $\mathbf{u} \cdot d\mathbf{\Sigma}$ from the 3-dimensional case considered earlier.

Hence, for closed volume Ω in phase space,

$$\int_{\partial\Omega} \omega = \int_\Omega d\omega = 0 \tag{A.6}$$

from which Eq. (A.2) directly follows.

The variation of action is therefore

$$\delta S = \int_{\partial\Sigma} \theta = \int_\Sigma d\theta = -\int_\Sigma \omega \tag{A.7}$$

for a trajectory along $\mathbf{u}$

$$\omega = 0 \tag{A.8}$$

thus,

$$\delta S = 0 \tag{A.9}$$